

CIRCULAR QUEUE

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

#define max 5

int front,rear,q[max];

void enqueue();

void dequeue();

void qdisplay();

void main()

{

    int c;

    clrscr();

    front=rear=-1;

    do

    {

        printf("\n choose:- 1.Enqueue\t2.Dequeue\t3.Display\t4.Exit\nEnter your choice:");

        scanf("%d",&c);

        switch(c)

        {

            case 1:enqueue();break;

            case 2:dequeue();break;

            case 3:qdisplay();break;

            case 4:printf("\n End");break;

            default:printf("\n Wrong choice");break;
```

```

    }
}while(c!=4);

getch();
}

void enqueue()
{
    int x;

    if((front==0&&rear==max-1)||((front==rear+1))

    {

        printf("\n Queue overflow");

    }

    if(front==-1)

    {

        front=rear=0;

    }

    else

    {

        if(rear==max-1)

        {

            rear=0;

        }

        else

        {

            rear++;

        }

    }
}

```

```
printf("\n Enter the number:");

scanf("%d",&x);

q[rear]=x;
}

void dequeue()
{
    int y;
    if(front==-1)
    {
        printf("\n Q is underflow");
    }
    y=q[front];
    if(front==rear)
    {
        front=rear=-1;
    }
    else
    {
        if(front==max-1)
        {
            front=0;
        }
        else
        {
            front++;
        }
    }
}
```

```
    printf("%d Successfully deleted",y);
}
void qdisplay()
{
    int i;
    if(front==rear==-1)
    {
        printf(" Q is empty");
    }
    printf("\n Elements are...");
    for(i=front;i!=rear;i=(i+1)%max)
    {
        printf("%d\n",q[i]);
    }
    printf("%d \n",q[rear]);
}
```